

New York City Sustainability Planning Assessment

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Introduction

New York City (NYC) is one of the most complex urban environments in the United States and the world, spanning a densely populated, diverse, and interconnected space. The city's vastness leaves it vulnerable to climate, environmental, and social risks. Over the course of the semester, our team has done extensive research and information gathering on NYC's sustainability, resilience, and socio-economic structures. Through class discussions, research on disaster scenarios, city plans, and environmental justice, we learned about NYC's city system. The city acts as a hub for food, culture, and finance; however, it faces challenges due to aging infrastructure, social inequalities, climate pressure, and resource allocation. NYC has many sustainability issues that come from the geographic landscape, social inequities, and the aging municipalities and infrastructure. The city spans around 520 miles of shoreline and faces threats to sustainability from coastal flooding, extreme heat disparities, hurricanes, and storm surges. ("NYC Comprehensive Waterfront Plan") With the growing population in the city, there is a high level of inequality in resources and housing. These inequities are prevalent in the environmental burdens that showcase how different boroughs, and particularly marginalized areas, face climate threats. Many of the buildings and infrastructure are over 70 years old (Miranda et al.), and various issues with transportation, municipal facilities, and stormwater management limit NYC's resilience and long-term planning. These factors all contribute to making NYC a place where sustainability planning must cautiously address adaptation, mitigation, resilience, and social and environmental equity while operating within a tight budget and extensive policies.

The term assignment had us draw on different types of data to understand how sustainability operates within a city. We learned to consider various ranges of limitations and opportunities regarding city planning for sustainability. We were able to draw information from lectures and data collected from workshops to draw conclusions and formulate a final memo. We were tasked with understanding how well cities plan and prepare for future risks related to climate and sustainability. Our assessment draws from the City Profile, the Sustainability Memo, and the presentations we did throughout the semester. We also draw upon each workshop for the respective assignments. Throughout these documents, we did extensive research regarding adaptation, mitigation, demographics, economics, governance systems, and disaster history. We referenced city plans and various websites related to NYC's structure and ecosystem. All of our research contributed to our evaluation of NYC's planning strategies, background, and sustainability initiatives.

Our main findings revolve around how NYC has strong long-term sustainability goals and ambitious climate agendas; however, it faces significant challenges regarding its infrastructure and governance. We found that plans such as OneNYC 2050 and the NYC Climate Adaptation Plan showcase key metrics relating to emissions, green infrastructure, resilience initiatives, and equitable investments and programs. We discovered prevalent sustainability and climate-related issues, such as flooding and high heat levels. We found there are significant gaps in coordination between agencies, inequitable distribution of resources to each borough, a lackluster infrastructure, low funding, and socioeconomic disparities. Our recommendations would be expanding green infrastructure, allocating funding for building improvements in marginalized zones, expanding resilience initiatives such as wetland restoration, and funding

flood-resistant green technology and infrastructure to strengthen adaptation, resilience, and sustainability.

This paper is organized into four sections. The first section focuses on the concerns regarding climate and sustainability based on NYC's background, demographics, history, governance, economics, and infrastructure. It also discusses policies and goals NYC has to address issues with detailed recommendations and proposals. The next section analyzes NYC's sustainability plans, focusing on how they impact prioritization, equity, nature, and their strengths and weaknesses. The following section focuses on policy recommendations to improve NYC's sustainability goals. Through this section, recommendations are specific to the key needs of the city and explore the impact of said policies. Lastly, the conclusion summarizes NYC's narrative, sustainability focuses, background, and policies.

Background & Context

Cities face mounting risks from climate change, including sea-level rise, extreme weather events, heat waves, flooding, and infrastructure disruption. Urban areas are particularly vulnerable because they concentrate people, economic activity, and critical infrastructure. Cities develop sustainability, climate action, and resilience plans to protect residents and infrastructure from climate impacts, meet state and federal emissions reduction targets, create green jobs while reducing long-term operational costs, address environmental justice and pollution-related health disparities, and demonstrate solutions that can be replicated globally.

Climate mitigation focuses on reducing greenhouse gas emissions to slow or prevent further climate change. For NYC, this primarily means addressing building emissions, which account for approximately 70% of citywide emissions ("Emissions"). Climate adaptation involves adjusting systems, processes, and infrastructure to moderate harm from current and

future climate impacts. This is reactive, accepting that some climate change is inevitable and preparing for its effects. Climate resilience encompasses the ability to anticipate, prepare for, respond to, and recover from climate disruptions while maintaining essential functions. It combines adaptation with the capacity to bounce back stronger after disasters.

Cities are pursuing mitigation by mandating emissions reductions for large buildings, transitioning to renewable energy and clean electricity, electrifying building heating systems and vehicle fleets, implementing zero-waste strategies and circular economy approaches, and expanding public transit and active transportation infrastructure. Adaptation actions include elevating critical infrastructure and mechanical systems above flood levels, installing green infrastructure for stormwater management, creating cooling centers and heat emergency response protocols, updating building codes for climate-informed design, and implementing early warning systems for extreme weather. Resilience strategies involve hardening coastal defenses and constructing storm surge barriers, diversifying energy and water supply systems, building redundancy into transportation networks, strengthening emergency response coordination, and investing in community-based preparedness programs.

New York City comprises five boroughs across approximately 300 square miles at the confluence of the Hudson River, East River, and Atlantic Ocean. The city has over 600 miles of coastline, making it vulnerable to coastal flooding and storm surge. NYC sits at the boundary between humid subtropical and humid continental climate zones, with average temperatures rising steadily since 1900. The Urban Heat Island effect can cause the city to be up to 22 degrees warmer than nearby suburban and rural areas.

Hurricane Sandy in 2012 remains the defining climate disaster for modern NYC. The storm killed 44 NYC residents with estimated damages of \$19 billion ("Hurricane Sandy"). The

subway system was flooded with eight tunnels submerged, and the MTA incurred about \$5 billion in damage. Approximately 800,000 people lost power, with some in the hardest hit areas living without electricity for months. Close to 50,000 homes in the Coney Island area were made uninhabitable, and close to half of all residents who died during the storm were over 65 years old, highlighting the vulnerability of elderly populations.

NYC faces ongoing climate risks from coastal flooding. By 2100, almost 50% of buildings in Lower Manhattan will be at risk from storm surge, and 20% of streets will be at risk of daily flooding ("Lower Manhattan Coastal Resiliency"). Extreme heat is the leading cause of weather-related deaths, disproportionately affecting seniors and those without adequate cooling. Heavy precipitation continues to overwhelm stormwater systems and cause localized flooding, while infrastructure designed for historical rather than projected conditions remains vulnerable.

NYC operates under a strong mayor-council system, with Mayor Adams creating the Mayor's Office of Climate and Environmental Justice (MOCEJ) in 2022 to consolidate climate authority and coordinate across multiple city agencies. The city faces stark economic disparities that intersect with climate vulnerability, with 18.2% of New Yorkers living in poverty in 2023 and significant racial wealth gaps. Median wealth is \$2,800 for Black New Yorkers and \$0 for Latinx New Yorkers, compared to substantially higher wealth for white residents. Environmental justice concerns are central to planning. Low-income communities and communities of color have borne disproportionate burdens from pollution, often live in flood-prone areas, and lack resources for adaptation. NYC aims to reduce PM2.5 air pollution disparities by 20% by 2030 ("Cities100: New York City"). NYC's infrastructure presents both critical assets and vulnerabilities, including a subway system with 472 stations serving over 1 billion annual riders that requires an estimated \$6 billion in climate resilience investments over the next decade, aging

combined sewer systems vulnerable to overflow, and nearly 50,000 buildings covered by emissions regulations as the energy sector accounts for approximately 70% of the city's greenhouse gas emissions ("Emissions").

OneNYC 2050 was released in 2019 as part of NYC's Green New Deal and comprises eight primary goals and 30 strategic initiatives for the next 30 years. It commits to executing a \$20 billion resiliency plan to address coastal storms, sea-level rise, extreme heat, and increased precipitation, with a commitment to carbon neutrality by 2050 ("OneNYC 2050"). The plan features four livable climate initiatives focusing on carbon neutrality, infrastructure resilience, economic opportunities, and climate justice. The 80×50 goal was codified into law by the City Council in 2015, mandating 80% greenhouse gas emissions reduction by 2050. NYC became the first city to map a local strategy to the UN Sustainable Development Goals ("New York City Aligns 2050 Strategy with SDGs"). The plan builds on PlaNYC of 2007 and 2011 and OneNYC of 2015, which established the foundation for environmental planning focused on equity.

The NYC Stormwater Resiliency Plan of 2021 represents the city's first comprehensive citywide analysis of flooding caused by extreme rainfall events. Unlike coastal storms, extreme rainfall can cause flooding in any part of the city, including inland neighborhoods miles from the shoreline. The plan outlines goals and initiatives to implement over 10 years, including new policies for resilient stormwater management, integration of future-looking climate change projections into long-term drainage planning, changes to flash flood emergency response procedures, and increased focus on public communications related to rainfall-based flooding ("Stormwater Resiliency Plan"). The New York City Panel on Climate Change has projected rainfall could increase by as much as 25% by the end of the century, placing a growing strain on drainage infrastructure that includes traditional sewer systems, nature-based Bluebelts, and more

than 10,000 distributed green infrastructure assets. The plan produced NYC's first stormwater flood maps showing vulnerability from rainfall events under future climate conditions.

The Draft 2026 Solid Waste Management Plan (SWMP26) provides a framework for waste reduction, reuse, recycling, and resource recovery over the next decade. The plan's overarching goal is to reduce waste across all generators and streams, expand and increase the efficiency of recycling and resource recovery efforts, improve existing waste management systems, responsibly manage special waste and reduce pollution and toxicity, and reduce greenhouse gas emissions that contribute to climate change ("Draft 2026 Solid Waste Management Plan," 1). New York City residents and businesses generate over 2 million tons of organic waste per year, with most discarded as refuse. The Citywide Residential Organics Program launched in 2024, expanding prior opt-in programs and aiming to increase diversion of organic waste through composting, anaerobic digestion, and codigestion of wastewater and separated organics. The plan builds on the 2006 SWMP and aligns with New York State's 2023 Solid Waste Management Plan, which sets forth six major focus areas to advance the circular economy, including waste reduction and reuse, recycling market development, product stewardship, and extended producer responsibility.

OneNYC 2050 prioritizes carbon neutrality by 2050, infrastructure resilience, economic opportunities, and climate justice, identifying coastal storms, sea-level rise, extreme heat, and increased precipitation as critical threats. The Stormwater Resiliency Plan addresses the previously overlooked issue of inland flooding from extreme rainfall, which can affect any neighborhood regardless of coastal proximity, with projections showing potential 25% rainfall increases by century's end, straining drainage infrastructure. SWMP26 focuses on waste reduction across all streams, expanding recycling and resource recovery, and reducing

greenhouse gas emissions from the 2 million tons of annual organic waste currently sent to landfills. These plans embody adaptation through infrastructure upgrades and flood mapping, mitigation via the 80×50 emissions reduction target and organic waste diversion, and resilience by building capacity to withstand multiple climate stressors while addressing equity concerns.

The plans recommend substantial capital investments of \$20 billion, regulatory changes to stormwater management policies and emergency response procedures, infrastructure development with green infrastructure assets, Bluebelts, and upgraded drainage systems, programmatic expansions such as the Citywide Residential Organics Program launched in 2024, and systemic reforms aligned with state and international frameworks including the state's circular economy focus and UN Sustainable Development Goals. They emphasize nature-based solutions alongside traditional infrastructure, integrate future climate projections into long-term planning, and explicitly prioritize environmental justice to ensure vulnerable communities benefit equitably from climate investments.

Sustainability Planning Analysis

Cumulatively, New York City's sustainability plans strongly and equitably address the most pressing climate threats of flooding and coastal storms, but inadequately address a lower-risk threat that disproportionately impacts elderly and low-income populations: extreme heat. Roughly half of NYC's hazard mitigation efforts undertaken in 2024 addressed flooding and coastal storms, whereas only 7.1% addressed extreme heat ("NYC Hazard Mitigation Plan"). Mitigation efforts focus on coastal storms and flooding because they pose a greater risk to the city as a whole. The city estimates hazard risks based on exposure and vulnerability, with exposure referring to how physically prone an area is to a hazard and vulnerability referring to the degree to which a hazard harms people in that area physically and economically. The Federal

Emergency Management Agency (FEMA) National Risk Index gives Manhattan a heat wave risk score of 99.1 and a coastal flooding risk score of 96.4, reflecting the high frequency, or exposure, of the city to heat waves (FEMA). However, NYC's previous experience with hazards has demonstrated its high vulnerability to flooding and coastal storms. Hurricane Sandy cost the city billions of dollars in damage to roads and the Subway system, the electric grid, and critical housing infrastructure ("Hurricane Sandy"). Additionally, NYC's low-income communities are the most exposed and susceptible to damage from flooding and coastal storms, as they tend to live in floodplains and lack the mobility to move or evacuate in response to a disaster ("Hurricane Sandy"). The Stormwater Resilience Plan addresses this vulnerability with changes to emergency response procedures, flood vulnerability mapping, and infrastructure upgrades targeting low-income communities in floodplains ("Stormwater Resiliency Plan," 14). Extreme heat, on the other hand, is only indirectly addressed by the city's waste and emissions reduction goals outlined in SWMP2026 and OneNYC 2050, with no focus on disproportionately vulnerable populations. Reducing emissions from waste and energy use may slow global climate change and the rising frequency and severity of heat events it produces, but it will not help elderly and low-income residents adapt to or rebound from heat waves. While NYC's high vulnerability to coastal storms and flooding demands attention, NYC's plans overlook the high exposure and vulnerability of elderly and low-income residents to extreme heat.

Although NYC's plans vary in how much they incorporate equity, together they lessen disparities in the distribution of flooding costs and adaptation benefits. OneNYC is the most justice-forward, seeking to improve social equity by improving economic opportunity and environmental conditions within vulnerable communities. OneNYC uses foresight to plan for a desired future in which communities that are vulnerable now have the economic strength to be

more resilient to climate shocks. OneNYC plans to attain a more inclusive economy in the future by creating good-paying jobs with fair wages and expanding benefits, ensuring that low-income communities benefit from adaptation (“OneNYC 2050”, 6). The Hazard Mitigation Plan also tries to ensure equitable distribution of benefits, such as the decarbonization of household electricity in affordable housing communities (“Hazard Mitigation Plan”). Meanwhile, the Stormwater Resiliency Plan incorporates historic disenfranchisement into its criteria for determining vulnerability to flooding, recognizing how low-income, low-elevation, coastal communities have borne the brunt of flood damages and making strides toward reducing those costs (“Stormwater Resiliency Plan,” 14). As a whole, these plans make progress toward a more equitable distribution of hazard costs and adaptation benefits, although by neglecting to address extreme heat, they still could do more to address disproportionate impacts on elderly and low-income residents.

Nevertheless, NYC’s plans align well with the Planner’s Triangle, a concept that positions sustainable development at the center of conflicting social, economic, and environmental priorities. NYC’s plans confront those conflicts by pursuing equitable economic growth. An inclusive, resilient economy with more well-paying jobs benefits not only low-income populations but also overall prosperity by generating and circulating more income, enhancing the ability of vulnerable communities to adapt and respond to climate hazards without municipal support. By investing present resources into equitable development, NYC will prevent greater conflicts over property, resources, and development due to the inability of communities to recover from anticipated climate shocks. The main areas of improvement for NYC’s plans to better reconcile conflicting priorities are by investing in infrastructure to reduce extreme heat risk. Green infrastructure supports social justice by cooling densely developed neighborhoods,

promotes economic growth by making low-income communities more energy-independent, and protects the environment by reducing carbon emissions from air conditioning.

New York City's planning increasingly integrates nature-based solutions to address climate resilience and stormwater management. The Stormwater Resiliency Plan highlights the use of "nature-based Bluebelts" and "more than 10,000 distributed green infrastructure assets" to manage rainfall and reduce strain on traditional sewer systems ("Stormwater Resiliency Plan," 5). Furthermore, OneNYC 2050 emphasizes nature-based solutions alongside traditional infrastructure to enhance resilience against coastal storms and flooding ("OneNYC 2050," 24). The city is also pursuing a "circular economy" approach through SWMP26, which aims to divert organic waste from landfills through composting and anaerobic digestion, thereby returning nutrients to the soil and reducing emissions ("Draft 2026 Solid Waste Management Plan," 5-6).

NYC's approach exhibits strong elements of urban metabolism and ecosystem services. The city's focus on waste reduction and the circular economy reflects an urban metabolism framework, where waste streams are minimized, and resources are recovered ("Draft 2026 Solid Waste Management Plan," 5-6). The utilization of wetlands and green infrastructure to absorb storm surge and manage stormwater demonstrates a reliance on ecosystem services for flood protection ("Stormwater Resiliency Plan," 18). Additionally, the integration of natural systems with built infrastructure, like the subway and sewer system, and social equity goals like protecting vulnerable communities, aligns with a Social-Ecological-Technological Systems (SETS) approach. For instance, OneNYC 2050 seeks to reconcile social, economic, and environmental priorities by investing in infrastructure that supports both environmental goals and social justice (OneNYC 2050, 3).

A major strength of NYC's plans is the comprehensive and equitable approach to addressing flooding and coastal storms. OneNYC 2050 and the Stormwater Resiliency Plan strongly prioritize these high-exposure threats, utilizing detailed flood mapping and infrastructure upgrades to protect vulnerable low-income communities ("OneNYC 2050," 3; "Stormwater Resiliency Plan," 14). However, a significant weakness is the inadequate attention to extreme heat. Despite extreme heat being a leading cause of weather-related deaths, only "7.1% of hazard mitigation efforts" address it compared to roughly half addressing flooding ("NYC Hazard Mitigation Plan"). The city's plans lack specific, robust adaptation strategies for heat that are comparable to its flood defense measures.

The plans address the issue of flooding exceptionally well, with the "Stormwater Resiliency Plan" specifically targeting the previously overlooked use of inland flooding from extreme rainfall and acknowledging the vulnerability of low-lying neighborhoods (Stormwater Resiliency Plan). In contrast, the issue of extreme heat is not addressed well. While emissions reductions may mitigate long-term climate change, they do not help residents adapt to immediate heat waves. The current plans overlook the high vulnerability of elderly and low-income residents to extreme heat, failing to provide sufficient cooling infrastructure or adaptation strategies for these groups ("NYC Hazard Mitigation Plan").

NYC is limited budgetarily and infrastructurally. The city relies on aging infrastructure, such as a subway system and sewer network that are over 70 years old, which requires massive capital investment for resilience upgrades, estimated at "\$6 billion... over the next decade" for the subway alone (Forman). Budgetary constraints also force the city to prioritize immediate, high-damage threats like coastal storms over 'lower risk' but deadly threats like heat. Additionally, the city operates within a complex political landscape involving multiple agencies,

where gaps in the coordination between agencies can hinder the implementation of comprehensive sustainability initiatives.

The most glaring gap is in heat adaptation. While flood resiliency is well funded and mapped, there is a lack of specific green roof initiatives and installation programs for heat-vulnerable boroughs. What is missing is a funded, robust strategy to deploy cooling infrastructure, such as green roofs and expanded tree canopies, to mitigate the urban heat island effect and protect vulnerable populations.

Policy Recommendations

New York City should expand green roof initiatives and installation programs in boroughs that face high levels of heat vulnerability, such as the Bronx, Brooklyn, and Queens. These areas are home to neighborhoods that are experiencing the Urban Heat Island effect as a result of the dense infrastructure, lack of green spaces and trees, and older, less heat-resistant buildings. High levels of heat exposure affect low-income residents and communities of color most, as they have less access to air conditioning and energy-efficient infrastructure (“Mayor’s Office of Climate & Environmental Justice”). Green roofs would reduce surface and air temperatures, improve air quality, and absorb stormwater, supporting less energy expenditures and advancing environmental justice.

NYC is characterized by old infrastructure, and we recommend that the New York City Housing Authority (NYCHA) allocate more funding for building improvement and redesign projects. This would include installing heat pumps, updating insulation layers, adding solar panels, and crafting green infrastructure on and around buildings. We learned how NYCHA housing developments are among the most vulnerable zones in the city, where residents face old systems, improper energy use, and high exposure to heat and flooding. Most NYCHA buildings

were built before modern building standards and face millions of dollars in construction and improvement needs (“Physical Needs Assessment Final Report,” 13). Upgrading the NYCHA buildings would improve air quality and reduce utility costs while reducing greenhouse gas emissions, increasing the overall resilience of the areas by reducing the reliance on fossil fuels.

NYC should focus on restoring and expanding the coastal wetlands to flood-prone boroughs such as Staten Island, Brooklyn, and Queens (“A Guide to Understanding NYC Flood Zones”). NYC’s wetlands act as natural protectors that absorb high levels of storm surge and reduce coastal flooding. The city’s wetland levels have been steadily receding over time due to land development, which increases levels of vulnerability to heavy rainfall storms and high sea levels (“Multi-Resolution Land Characteristics Consortium”). Nature-based solutions, such as building seawalls, are the most cost-effective way to handle these issues. Restoring the wetlands would provide a long-term solution to flooding and reduce future damage to communities while ensuring costs are minimized and recovery is faster when major storms hit. Wetlands would improve water quality and biodiversity while pushing forward resilience and sustainability goals (“NYC Housing Recovery”).

Lastly, we suggest that NYC expand funding and grants through public and private means to provide green infrastructure to fight sewer overflows and poor water quality. This would be through the implementation of bioswales, permeable pavement, and rain gardens in public and private zones to reduce flooding in vulnerable neighborhoods such as the Bronx, Queens, and Brooklyn. These boroughs have old sewer systems and face flooding due to the construction of impervious surfaces. The Stormwater Resiliency Plan acknowledges this issue; however, the implementation is not at the desired scale to achieve resilience and meet sustainability goals. This is primarily due to the lack of funding and coordination. Increasing funding to provide

green infrastructure would reduce sewer overflows and improve water quality while also lowering the costs of repair and maintenance. Focusing on providing equal treatment to all the boroughs will ensure that equity goals are met while addressing infrastructure issues.

Conclusion

New York City stands at a critical juncture in its sustainability journey, balancing ambitious long-term goals against the immediate realities of climate change and aging infrastructure. As this memo has outlined, the city faces compounding risks from coastal storms, inland flooding, and extreme heat, threats that are worsened by deep-seated socioeconomic disparities and a built environment that is largely over 70 years old. Through OneNYC 2050 and the Stormwater Resiliency Plan, the city has established a strong foundation for resilience, using nature-based solutions such as Bluebelts and committing to equitable growth that prioritizes vulnerable communities. However, our analysis reveals a significant gap between the city's robust flood defense strategies and its inadequate response to extreme heat. While the city has effectively integrated ecosystem services and urban metabolism frameworks, evident in its circular economy goals for waste management, it has not sufficiently funded adaptation strategies for the urban heat island effect, which disproportionately impacts elderly and low-income residents. Furthermore, the implementation of these sustainability goals is frequently hampered by budgetary constraints and the sheer scale of required capital investments for public assets like the subway system and sewers. To bridge these gaps and achieve a true balance of social, economic, and environmental priorities, New York City must pursue more targeted, community-level interventions. We recommend that the city expand green roof initiatives in heat-vulnerable boroughs, allocate specific funding for energy-efficient retrofits in NYCHA housing, actively restore coastal wetlands, and aggressively implement green infrastructure to

manage sewer overflows. By pivoting resources toward these high-impact, equity-focused solutions, New York City can move beyond mitigation and ensure a resilient, habitable future for all its citizens.

Citations

“A Guide to Understanding NYC Flood Zones.” CityRealty,

www.cityrealty.com/nyc/market-insight/resources-and-guides/flood-zones.

“Cities100: New York City.” C40 Cities,

<https://www.c40.org/case-studies/cities100-new-york-city-planning-for-an-equitable-and-sustainable-city/>.

"Draft 2026 Solid Waste Management Plan." NYC Department of Sanitation, City of New York,

www.nyc.gov/assets/dsny/downloads/resources/reports/solid-waste-management/2026-swmp/draft-swmp-2026.pdf.

"Emissions." Office of the New York City Comptroller Brad Lander,

comptroller.nyc.gov/services/for-the-public/nyc-climate-dashboard/emissions/.

Forman, Adam. “NYC’s Silent Infrastructure Challenge: Aging Public Buildings.” Center for an Urban Future (CUF), 2024,

nycfuture.org/research/city-limits-nycs-silent-infrastructure-challenge-aging-public-buildings.

"Hurricane Sandy." Community Development Block Grant - Disaster Recovery, City of New

York, www.nyc.gov/site/cdbgdr/hurricane-sandy/hurricane-sandy.page.

“Lower Manhattan Coastal Resiliency.” New York City Economic Development Corporation,

<https://edc.nyc/project/lower-manhattan-coastal-resiliency>.

“Mayor’s Office of Climate & Environmental Justice.” *Nyc.gov*, 2019,

www.nyc.gov/content/climate/pages/.

“NYC Housing Recovery.” Mayor’s Office of Recovery and Resiliency, 2024,

www.nyc.gov/site/housingrecovery/resources/mayors-office-of-resiliency-mor.page.

Miranda, Leo et al. "Are You Traveling Over Troubled NYC Bridges?" *The City*, 2025,
<https://www.thecity.nyc/2025/05/20/new-york-crumbling-old-bridges/>.

"Multi-Resolution Land Characteristics Consortium." *Mrlc.gov*, 22 Aug. 2024,
www.mrlc.gov/home.

"New York City Aligns 2050 Strategy with SDGs." International Institute for Sustainable
Development, <https://sdg.iisd.org/news/new-york-city-aligns-2050-strategy-with-sdgs/>

"NYC Comprehensive Waterfront Plan." Department of City Planning, 2021,
[https://www.nyc.gov/assets/planning/download/pdf/plans-studies/comprehensive-waterfr
ont-plan/nyc_comprehensive_waterfront_plan_lo-res.pdf](https://www.nyc.gov/assets/planning/download/pdf/plans-studies/comprehensive-waterfront-plan/nyc_comprehensive_waterfront_plan_lo-res.pdf).

"NYC Hazard Mitigation Plan." New York City Emergency Management, 2024,
<https://nychazardmitigation.com/documentation/nyc-hazard-mitigation-plan/>.

"OneNYC 2050." NYC Mayor's Office of Climate and Environmental Justice, City of New
York,
https://www.nyc.gov/assets/climate/downloads/pdfs/OneNYC_2050_Strategic_Plan.pdf.

"Physical Needs Assessment Final Report." New York City Housing Authority, 2023,
[https://www.nyc.gov/assets/nycha/downloads/pdf/2023-PNA-Report-Physical-Needs-Ass
essment-NYCHA.pdf](https://www.nyc.gov/assets/nycha/downloads/pdf/2023-PNA-Report-Physical-Needs-Assessment-NYCHA.pdf).

"Resilience Analysis and Planning Tool." Federal Emergency Management Agency,
<https://experience.arcgis.com/experience/0a317e8998534c30a9b2d3861c814d42/>.

"Stormwater Resiliency Plan." NYC Water, City of New York Department of Environmental
Protection, 6 Jan. 2022,
<https://www.nyc.gov/assets/orr/pdf/publications/stormwater-resiliency-plan.pdf>.

"To Reduce Flooding and Improve the Health of Waterways, New Rule Enhances Stormwater Management." NYC Department of Environmental Protection, City of New York, 11 Jan.

2022,

www.nyc.gov/site/dep/news/22-006/to-reduce-flooding-improve-health-waterways-new-rule-enhances-stormwater-management.